(A) Training on Build2SPARQL (154 train buildings)

backpropagation: update only the orange (trainable) modules

INPUT

Training sample
- question q
- gold SPARQL Q*
- building ID

Building KG G = (V, R, E)
Brick or ASHRAE 223P
(instances + class hierarchy)

Text encoder (frozen)
q → d-dim question vector q

Seed linking
mentions/classes → seeds S ⊂ V

Relation graph G_R
nodes = relation types
edges = 4 meta-relations

edges E (message passing graph)

Entity GNN (NBFNet)
conditional message passing
$h^0(v) = 1[v \in S] \cdot W_0 q$
$h^t(v) = AGG(h^{t-1}(u) \odot w_r)$, L layers
→ question-aware states h_q(v)

Relation GNN (ULTRA)
→ relation embeddings {w_r}
(works for unseen relations)

Scoring heads
node score s(v)
edge score s(u, r, v)
path score Σ log s(edge)
(GNN OUTPUT)

Loss
BCE(s(v), Y_V)
+ BCE(s(u,r,v), Y_E)
+ λ · contrastive(P*, sibling paths)
compare prediction vs. labels

Q* and G (for label generation)

Automatic label generation (offline, no manual annotation)
execute gold SPARQL Q* on G → bind all triple-pattern variables →
evidence nodes Y_V · evidence edges Y_E · gold class-level path P*
(e.g. VAV –feeds→ HVAC_Zone –hasPoint→ Temp_Sensor)

labels

(B) Inference on unseen buildings (Build2SPARQL test · BuildingQA · Stanford case study)

INPUT

NL question q
"Which temperature sensors are in zones served by AHU-1?"

Unseen building KG G
never seen in training
(Brick or 223P)

Trained BRIDGE retriever
(weights fixed after training)
text encoder + seed linking
+ relation GNN + entity GNN
+ scoring heads

Retrieved context
top-k nodes / edges
→ beam search → top-k paths
→ class-level, verbalized:
AHU –feeds→ VAV –feeds→ Zone …

LLM agent (BrickQA)
write SPARQL from paths
validate (syntax, path, arity) → execute → revise
(not trained)

OUTPUT

SPARQL query + answer table
evaluated by
Row-Matching F1
(retriever: Recall@k)

execute SPARQL against the building KG

data / input · fixed (not trained) · trainable GNN · training objective · LLM (frozen, prompted) · output · gradient flow

GNN input: (q, G, S) → GNN output: node/edge/path scores → system output: SPARQL + answer table